ADITYA NARAYAN SINGH

AI Engineer

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Summary

AI/ML postgraduate with 3 years of experience building production-grade AI and data solutions. Expert in architecting scalable ML pipelines, feature engineering workflows, and agentic AI systems using Python, FastAPI, and Google Cloud (Vertex AI, BigQuery, Dataproc). Skilled in prompt engineering, LLM fine-tuning, and RAG pipeline development with LangChain and Hugging Face.

Education

MSc. In Artificial Intelligence and Machine Learning **09/2025 – 09/2026**

University of Birmingham, Edgbaston

- Key Modules: Mathematics, Machine Learning, Artificial Intelligence, Computer Vision & Imaging, NLP

B.Tech. in Computer Science & Engineering **08/2018 – 08/2022**

Dr. APJ Abdul Kalam Technical University, India

- CGPA: 7.58
- *Key Modules: Data Structures, Algorithm Design, Database Management Systems, AI/ML, Data Analytics*

Skills

- **AI/GenAI:** Multi-agent Systems (Google ADK), RAG, LLMs, LangChain, LlamaIndex Hugging Face, NLP, Computer Vision, Prompt Engineering, Model Fine-tuning.
- **Cloud Engineering:** Google Cloud Platform (Vertex AI, Dataproc, BigQuery), Azure, MLOps
- **Tools & Frameworks:** Python, TensorFlow, PyTorch, FastAPI, PySpark, Docker, Kubernetes, MLflow, NiFi, DBT, GitLab, GitHub Actions, Airflow.
- **Strategy:** Technical Discovery, Stakeholder Management, Requirement Gathering, Mentoring.

Work Experience

Data Engineering Analyst **10/2022 – 07/2025**

Accenture Solutions Pvt. Ltd, Pune

- Architected and deployed end-to-end predictive pipelines using NiFi and GCP, reducing architecture latency by 40% and accelerating data-driven decision-making for high-value clients.
- Engineered and optimized ETL workflows for 300M+ records, cutting data-to-model latency by 30% to enable real-time predictive analytics.
- Led technical discovery and requirement gathering for strategic code improvements, ensuring AI solutions met rigorous business and infrastructure requirements.
- Delivered high-impact presentations to business stakeholders, translating complex ML enhancements and algorithm optimizations into measurable ROI.
- Completed professional certifications in Tableau, Informatica, PowerBI, DataProc, PySpark, Communication.

Project Intern **05/2020 – 06/2020**

Robust Results Pvt. Ltd., Kanpur

- Developed predictive analytics solutions using Python, Pandas, and NumPy, achieving 90% forecasting accuracy to support strategic risk assessment.
- Optimized ensembles of classification/regression models, ensuring production-ready performance across diverse enterprise datasets.

Publication

Self-Workout Trainer (May 2022): Published research paper for computer vision academic project which implemented skeletal posture analysis for training a model which gives warning when the user in frame is doing exercise with wrong posture.

Certifications

- Microsoft Certified **Azure AI Engineer Associate (12/2024)**
- Google Cloud Platform (GCP) **Professional Data Engineer (10/2024)**
- Artificial Intelligence Certification – E&ICT Academy, **IIT Kanpur (07/2022)**
- Neural Networks and Deep Learning by DeepLearning.AI **(08/2022)**
- Computer Vision Basics by University at Buffalo **(01/2021)**

Projects

- **AI ChatBot & LLM Integrations (2024 – Present):** Developed a personal AI assistant chatbot integrating LangChain and RAG pipelines to track workflows and class schedules. Successful implementation of multi-lingual NLP, Hugging Face, REST APIs.
- **Multi-Agent Support System (2024):** Deployed a 5-agent orchestration system using Google ADK across 7 task types, implementing tool-use, task routing, and agent-to-agent communication estimated to save 4 hours per week.
- **Virtual Dressing Room Project (2023):** Led a team to develop a deep learning model using TensorFlow and PyTorch, enhancing virtual try-on accuracy for fashion retail. Served via FastAPI and deployed using Docker and Kubernetes with CI/CD integration.
- **Defect Recognition Model (2023):** Trained a computer vision model for defect recognition in circular steel components using OpenCV, Javascript and pattern recognition, achieving 90% classification accuracy on 20 real-time test samples, served via a Flask REST API.
- Mask Detection, Pose estimation for workouts, Sentiment analysis, Toxic comments recognition models created and details available on GitHub link.

Achievements

- **HealthTech AI 2026 winner** (1st of 7 teams), delivering a Doppler signal-processing pipeline with ~80% noise reduction and 90% model accuracy using CNN, DSP pipeline with FastAPI backend, frame slicing, and sliding window algorithm.
- **FrontierTechX Loyalty Program Challenge winner** (1st of 11 teams), built a token-based reward ecosystem targeting MSMEs transaction growth, implemented end-to-end using Python, stable coin integration, and banking APIs.
- Received **recognition from the manager** for sustained high performance across multiple high-value client deliveries over a 3-year tenure at Accenture.

Extracurricular Activities

- Spearheaded the team in CVA 2020 with a homestay business model and improved medical device implementing critical thinking.
- Co-ordinated the volunteer team while associated with a dog NGO to rescue, feed and vaccinate dogs in the assigned region utilizing teamwork and leadership skills.
- Volunteered as campaign advisor, strategist, and social media manager for an International Officer candidate in the University of Birmingham Student Officer elections.
- Collaborated in a team during a Google hands-on workshop to solve a creative brief on Vertex AI, utilizing Gemini and Veo for product advertisement development.
- Volunteered as a coordinator and helper at the Indian themed event of friends international society.

Languages

- English – Fluent
- German – Basic
- Hindi – Native
- Punjabi – Advanced

References available on request.